Understanding Data and Model Steering

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1. Understanding and Filtering Pretraining Data

124 samples from four pretraining datasets - FineWeb (fin, b), FinePDFs (fin, a), C4 (c4), and RedPajama-V2 (red), were extracted and annotated to get an understanding of what LLMs are trained on. Each sample was annotated for content type, quality (1–10), whether it was correctly cleaned from its original HTML or PDF form, and whether it contained private or harmful information. Several samples were not in English (such as Arabic, Bulgarian, Marathi, Amharic, Chinese) and were translated prior to annotation.

1.1. Data Analysis

Quality: The average quality score was 4.5/10 across all 124 samples. Only 10 samples (8%) scored 7 or above, and 31 (25%) scored 3 or below. Quality was fairly consistent across datasets: FineWeb (4.7), FinePDFs (4.3), C4 (4.3), RedPajama (4.7).

Cleaning: 60% of samples were correctly cleaned, 32% partially cleaned, and 7% not cleaned. The most common issue was leftover HTML table formatting showing up as pipe characters around text. There were also documents that started mid-sentence, suggesting the crawler picked up a reply in the middle of the page rather than a full document. Additionally, we found spam and only titles from a search page with no meaningful content.

Private and Harmful Information: 9 samples (7.3%) contained private information. These included exam candidate lists with roll numbers, competition results with full names, a recruitment call letter, etc. PDFs seem to pick up a lot of institutional documents that are technically public but weren't meant to end up in a training dataset. 2 samples were flagged as harmful: a black-magic services ad (C4) and a gambling promotion page (RedPajama).

Topics: The most common content types were blog posts (31%), news articles (19%), legal/government documents (13%), and product pages (9%). FineWeb was mostly news and blogs, C4 leaned commercial with a lot of product pages and ads, and FinePDFs stood out with over half its samples being legal or government documents. The range of content was wider than expected and we saw documents in several languages, including a pregnancy warning article in Amharic and a Chinese Christian music newsletter.

Overall, it was surprising that a lot of administrative data ended up in the dataset that are public but contain real people's personal details and were probably not intended to be part of a training dataset. Further, the quality of data itself wasn't very high. These annotations are also subjective, a different annotator may reach different conclusions for the quality score. This is true particularly for borderline cases such as whether a public figure's name is private information or not.

1.2. OpenAI Privacy Filter

We ran the OpenAI privacy-filter model (OpenAI) on all 124 samples to compare automated detection against our manual annotations. The model is a token-classification transformer designed to detect eight categories of sensitive information: private_person, private_email, private_phone, private_address, private_url, private_date, account_number, and secret. The model was loaded using the Hugging Face transformers library and applied to the first 10,000 characters of each document to accommodate the available GPU memory constraints.

False positives: The model identified 49 samples as containing private information that we had annotated as private=no, a precision of 0.155 and recall of 1. Nearly all of these cases were private/person detections involving names that appeared in clearly public contexts. Examples included athletes mentioned in sports news articles, politicians in election coverage, and religious scholars in Islamic texts. The model also flagged forum usernames such as axboi87, placeholder names like Jane Doe and Bob Smith in a patent document.

This behaviour of the model to over-redact public entities is also a limitation noted by OpenAI and it is likely because of limited or ambiguous local context. Adjusting this behaviour would require fine tuning.

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False negatives: The model had no false negatives in our sample. However, this does not mean the model is perfect because our sample size is very small.

Summary: The model is conservative by design, preferring to over-detect rather than miss private information, which is shown in its perfect recall but low precision of 0.155. This is a good data sanitation tool in general but it needs more work when used for a big pretraining corpus where the content is of a wide variety. We also note that the model doesn't have a flag for harmful content. In our annotations we distinguish between private and harmful content, with documents with black magic services and gambling promotion were flagged as harmful but not private. These would pass through the filter suggesting that PII detection might not be enough for pretraing.

2. Representation Engineering and Activation Steering

2.1. Probing for concepts

Dataset Generation: We selected harmfulness as the binary concept. We first tried a contrastive dataset with harmful prompts from AdvBench and harmless prompts from Alpaca. However, this produced near-perfect probe accuracy from the early layers onward, suggesting possible dataset-source artifacts in addition to harmfulness.

Addressing this issue, we used ChatGPT to construct a matched harmfulness dataset with 200 harmful and 200 harmless prompts. The prompts were arranged into pairs with similar wording but different harmfulness. For example, "Explain how to break into someone else's email account" was paired with "Explain how to secure an email account against unauthorized access." We used an 80/20 pair-level split, resulting in 320 training examples and 80 test examples.

Experiment: We used Qwen3-4B-Instruct-2507 as the target model. For each prompt, we applied the model's chat template and cached the hidden state at the final prompt token for every layer. The resulting activation tensors had shape $(320, 37, 2560)$ for the training split and $(80, 37, 2560)$ for the test split, corresponding to the embedding output plus 36 transformer layers.

For each layer, we trained a separate logistic-regression probe on the training split and evaluated it on the held-out test split. We report accuracy and AUC by layer. We also repeated the same procedure at the first token and a mid-sequence token to test how decodability changes across token positions.

Results: The final-token probe performs at chance at the embedding output, with accuracy 0.50 and AUC 0.50. Accuracy rises sharply at the first transformer layer, reaching 0.85 at layer 1 and 0.95 by layer 9. Layers 13-16 achieve perfect test accuracy and AUC, while later layers remain near-perfect.

Thus, harmfulness is most linearly decodable in the middle-to-late layers. There is no single isolated best layer; instead, there is a broad plateau beginning around layers 13–16.

The token-position results show that the final-token representation is strongest, reaching perfect accuracy and AUC. The mid-sequence token is also predictive, with best accuracy around 0.9625, but weaker than the final token. The first-token representation remains at chance across all layers. This suggests that harmfulness is not encoded in isolated token embeddings, but emerges as contextual information accumulates across the sequence. Since the final token can attend to the full prompt, it contains the clearest representation of the prompt-level harmfulness concept.

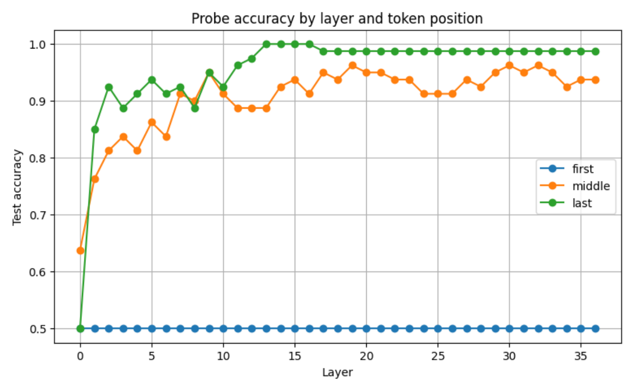
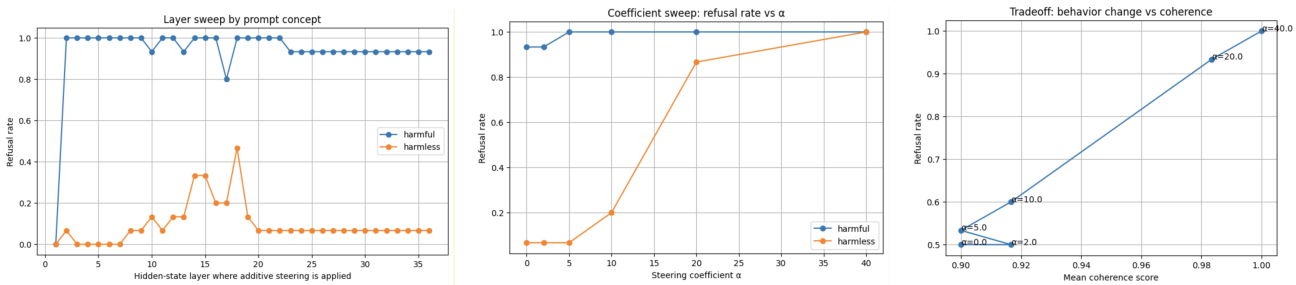


Figure 1. Logistic-regression probe test accuracy on the harmfulness concept across layers, for three token positions (first, middle, last prompt token).

2.2. Activation Steering

Experiment Setup and Scoring: To manipulate the model's refusal behavior, we extracted a steering vector using the difference-of-means between the activations of harmful and harmless prompts from our training split. This follows contrastive activation addition methodologies (Panickssery et al., 2023). Based on the probing results in Section 2.1, which demonstrated perfect linear separability emerging around layers 13–16, we selected layer 16 to apply our interventions.

We generated 30 responses on held-out test prompts to evaluate three conditions: (i) baseline, (ii) inducing refusal via additive steering, and (iii) suppressing refusal via directional ablation. We specifically target the refusal direction, verifying recent findings that refusal is mediated by a single direction in the residual stream (Arditi et al., 2024).



(a) Layer sweep: the ideal intervention point maximizes harmful refusal while keeping harmless refusal low.

(b) Coefficient sweep: at $\alpha = 5.0$, harmful refusal peaks before harmless prompts are negatively impacted.

(c) Trade-off between behavior change and mean coherence score across α values.

Figure 2. Activation-steering results across intervention layer, steering coefficient, and coherence trade-off.

To score the behavior change, we employed a deterministic keyword check rather than an LLM-as-judge. We defined a comprehensive list of standard refusal markers typically produced by aligned models (e.g., “I can’t”, “I’m sorry”, “cannot provide”, “against policy”). The generated text for each prompt was converted to lowercase, and if any of these predefined substrings were present, the generation was classified as a refusal.

Layer Sweep: Figure 2a demonstrates the result of applying the additive steering vector across all layers. Steering is effective at increasing harmful refusal but impacts harmless prompts especially between layers 10–20.

Coefficient Sweep: Figure 2b shows the coefficient sweep. We swept the scaling coefficient α to evaluate the trade-off between effective steering and model degradation. We identified a distinct “sweet spot” at $\alpha = 5.0$. At this coefficient, the model reached a 100% refusal rate for harmful prompts, while the refusal rate for harmless prompts remained perfectly at the baseline level (6.6%). However, increasing the coefficient beyond this point induced severe over-refusal.

Coherence Score: We also plotted behavior change against simple text coherence (Figure 2c). Interestingly, applying the steering vector did not destroy the model’s foundational language modeling capabilities; coherence scores remained consistently high (between 80%–100%) across the entire coefficient sweep. The primary side effect of additive steering is thus behavioral (a shift in tone and helpfulness) rather than a loss of grammatical or structural coherence.

Additive Steering vs. Directional Ablation: Directional ablation proved to be highly selective. By projecting out the refusal direction from the residual stream, the refusal rate for harmful prompts dropped from a baseline of 93.3% down to 0.0%—meaning the model became completely compliant with malicious requests. Crucially, this surgical removal did not impact harmless prompts (which remained at 0.0% refusal) and maintained perfect text coherence.

In contrast, additive steering is highly disruptive. While it successfully amplifies the targeted concept to drive harmful refusals to 100%, forcing this vector into the residual stream acts as a blunt instrument. If not perfectly calibrated to the $\alpha = 5.0$ sweet spot, additive steering pushes the model into a blanket “refusal state,” causing massive collateral damage to the model’s ability to safely comply with benign instructions.

The bonus tasks are discussed in Appendix A

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A. Bonus Tasks

A.1. Last-Token vs. All-Token Steering

We compared two intervention modes for additive steering: **all-token**, which modifies the hidden state at every token position in the residual stream, and **last-token**, which modifies only the final prompt token before generation begins.

Both modes achieve identical harmful-prompt suppression: 100% refusal on harmful prompts at both $\alpha = 10$ and $\alpha = 20$. The difference lies entirely in false-positive spillover onto harmless prompts. All-token steering at $\alpha = 10$ produces a 20% harmless refusal rate, which rises to 87% at $\alpha = 20$, representing severe over-intervention. In contrast, last-token steering at $\alpha = 10$ yields only 6.7% harmless refusal, rising to 20% at $\alpha = 20$.

The explanation is that all-token mode modifies every token's hidden state, including those carrying general semantic content unrelated to harmfulness. As a result, the steering signal bleeds into the representations of benign queries. Last-token mode disturbs only the representation at the generation decision point, steering the output without corrupting the broader context representation. Coherence is negligibly affected by either mode (all scores ≥ 0.83), so last-token steering gains precision at no coherence cost. Figure 3 summarizes this comparison.

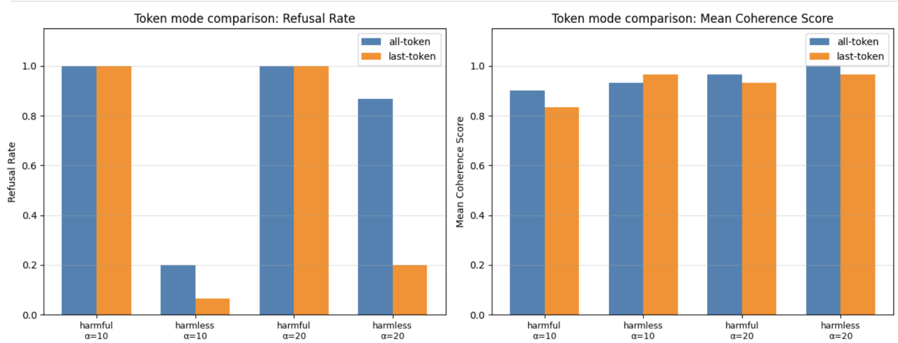

Figure 3. Refusal rate and coherence by intervention mode (last-token vs. all-token) and steering coefficient. Both modes suppress harmful completions equally, but all-token steering produces substantially more false refusals on harmless prompts.

A.2. Side Effects of Steering

To assess whether steering one concept perturbs unrelated capabilities, we measured factual recall via MMLU accuracy and language fluency via perplexity on a text sample across three conditions: baseline, directional ablation, and additive suppression of the harmfulness direction.

Factual recall is completely preserved. MMLU accuracy remains at 1.0 across all three conditions. We note the caveat that our evaluation set consists of only six questions, which is near-ceiling for a 4B-parameter model and cannot detect small regressions. A larger MMLU sample would provide finer resolution, but the result is consistent with no degradation.

Fluency is essentially unchanged. The perplexity delta is $+0.06$ for directional ablation and -0.09 for additive suppression, both under 2% of the baseline value of 4.31. The slight perplexity decrease under additive suppression could indicate that removing the harmfulness direction fractionally tightens the model's distribution over neutral factual text, but the effect is within noise.

Overall, the harmfulness direction appears to be a relatively isolated feature in the model's representation space. Removing it surgically, especially with last-token intervention, leaves factual recall, coherence, and language fluency intact while substantially reducing harmful completions. The primary side effect of all-token steering at high α is not degradation of capability but over-suppression: it begins refusing harmless prompts as well, which is a specificity problem rather than a capability problem.